HW05 Task 2 - AI Analysis and Misinterpretation Hunt

Student ID: 23127035
Run/analysis date: 2026-09-01
Workflow: WF-04 - Admin Login -> Import Product CSV -> Search Imported Product -> View Product Detail
AI tool: OpenAI Codex

1. Method and evidence boundary

The unreviewed AI response is retained in `audit/task2_first_pass_ai_analysis.md`. The correction below is reproduced from the four accepted `20260901` JTL files by `scripts/analyze_task2.py`; machine-readable output is `analysis/task2_metrics.json`.

Percentiles use nearest rank over JMeter `elapsed`. A row fails when JMeter `success` is not `true`, so business-assertion failures count even for HTTP 2xx. A workflow completes only when the successful final sampler `FR06 View imported product detail` exists. This is stricter than dividing samples by four because scheduled Load/Stress/Spike shutdown can leave partial iterations.

The technical review is complete but is not labelled as the student's human review until the student checks and approves or amends it.

2. Authoritative raw-log summary

Scenario	HTTP samples	Failed	p95 ms	p99 ms	Max ms	Samples/s	Completed workflows	Workflows/s
Load	1,912	0	5	6	72	16.04	477	4.00
Stress	6,514	0	4	5	19	36.33	1,613	9.00
Spike	3,216	0	4	5	50	26.98	793	6.65
Soak	49,200	0	4	6	1,151	79.49	12,300	19.87

Raw sources are `results/23127035_{Load,Stress,Spike,Soak}_20260901.jtl`. Overall, stage, sampler, and resource details are in `analysis/task2_metrics.md`.

3. Misinterpretation hunt

First-pass AI statement	Correct raw value	Why it is wrong
“Soak sustained 79.49 workflows/s.”	**79.49 HTTP samples/s**, exactly **12,300** final samplers and **19.87 completed workflows/s** over 618.94 s.	JMeter's default rate counts HTTP samplers, not business journeys.
“Stress achieved 36.33 requests/s,” used as its peak.	Whole Stress is 36.33 samples/s, but the **24-VU stage is 62.21 samples/s**, p95 4 ms, 0 failures.	The whole value averages 6-, 12-, and 24-VU windows.
“Spike is slower than Stress, so the burst degraded throughput.”	The **30-VU burst reached 84.03 samples/s**, p95 4 ms; 26.98 is the whole test including two 3-VU periods.	Different workload windows are not comparable peak stages.
“Spike did not recover.”	Recovery was **8.38 samples/s, p95 4 ms**, versus baseline **8.13 samples/s, p95 5 ms**.	Recovery must be compared with the same 3-VU baseline, not the burst or whole-run average.

First-pass AI statement	Correct raw value	Why it is wrong
“The 1,151 ms maximum proves a tail-latency failure.”	Soak p95 is **4 ms** and p99 **6 ms** across 49,200 samples; the maximum occurs in a brief multi-sampler cluster.	A maximum is an outlier, not a percentile or evidence of sustained tail degradation.
“0% errors proves production capacity at 79 users/s.”	Evidence proves **30 VUs**, 0 failures, 79.49 samples/s, and 19.87 workflows/s. No stage reached collapse.	VUs, user arrival rate, HTTP sample rate, and completed workflow rate are different quantities.
“Peak RSS 136.67 MB is the memory ceiling.”	Soak RSS start/peak/final is **67.64/136.67/51.86 MB**; first/last-quarter means are **111.26/51.41 MB**.	A transient peak followed by reclamation is neither an absolute ceiling nor proof of a leak.
“The suggested numbers are an SLA.”	Neither the assignment nor SUT defines a response-time/throughput SLA.	Local measurements support proposed regression gates only; product owners must approve service targets.

4. Proposed same-environment regression gates

Metric/scope	Proposed rule	Basis
Business correctness	Fail on any HTTP/business assertion failure.	All accepted samples passed; functional failures must not be averaged away.
Stable-stage p95	Warn above 7.5 ms; fail above 10 ms.	+50%/+100% over worst accepted p95 of 5 ms.
Stable-stage p99	Warn above 9 ms; fail above 12 ms.	+50%/+100% over worst accepted p99 of 6 ms.
Load 6 VUs	Fail below 14.44 samples/s.	10% below observed 16.04.
Stress 24 VUs	Fail below 55.99 samples/s.	10% below observed 62.21.
Spike 30-VU burst	Fail below 75.63 samples/s.	10% below observed 84.03.
Soak 30 VUs	Fail below 71.54 samples/s or 17.89 workflows/s.	10% below observed 79.49 and 19.87.
Backend resources	Investigate above 30% Node CPU or 165 MB RSS; do not auto-fail.	Rounded bands above measured 22.3% and 136.67 MB peaks; host and GC dependent.

These are candidate regression controls, not contractual SLAs. Repeated clean runs are required to estimate natural variance before CI enforcement or re-baselining.

5. AI recommendations judged against source

Recommendation	Classification	Source-grounded judgment
Wrap FR16 batch inserts in an explicit transaction.	**Feasible; prioritize experiment**	`server.js:209–240` prepares one statement but has no `BEGIN/COMMIT`. A transaction can reduce per-write commit overhead and make the batch atomic; benchmark multi-row imports and rollback behavior.
Enable SQLite WAL and `busy_timeout`.	**Feasible experiment, not guaranteed**	`database.js:5` opens one SQLite handle and defines no such PRAGMA. Measure writer contention and checkpoint behavior.
Add a non-unique index on `users(email)`.	**Feasible; profile first**	`database.js:50–61` has no email index; `server.js:35` performs equality login lookup. The local dataset has only 82 users and no observed login bottleneck.
Add a B-tree index on `products(name)` for current search.	**Hallucinated as a direct fix**	`server.js:144` uses `LIKE '%term%'`; the leading wildcard generally prevents ordinary prefix-index lookup. Examine `EXPLAIN QUERY PLAN`; consider FTS5 or a requirement change.

Recommendation	Classification	Source-grounded judgment
Add another index on `coupons(code)`.	**Hallucinated/redundant**	`database.js:31` already declares `code TEXT UNIQUE`, which creates a uniqueness index. Coupon lookup is not even part of WF-04.
Add a conventional database connection pool.	**Hallucinated for current architecture**	Current code uses one in-process SQLite handle, not a client/server database. A PostgreSQL/MySQL-style pool is not a drop-in fix and does not remove SQLite's single-writer constraint.
Cache exact product-search responses.	**Unsupported for this workload**	Every iteration imports a unique product then immediately reads it. Hit rate is near zero and invalidation is mandatory for read-after-write correctness.
Run clustered Node workers.	**Hallucinated as an immediate fix**	Multiple processes create multiple SQLite handles and may increase writer contention. This is an architectural experiment, not an evidenced fix for these results.

The first optimization experiment should compare multi-row FR16 imports with and without one explicit transaction, then test WAL/busy-timeout combinations. Query plans and representative data volume must precede index claims.

6. Bounded conclusion and review ownership

WF-04 remained stable inside the tested envelope: zero assertion failures, proportional Stress scaling, and Spike recovery to the 3-VU baseline. The accepted endurance result is 30 VUs for 618.94 seconds at 79.49 HTTP samples/s and 19.87 completed workflows/s. It is not 79 workflows/s, production readiness, absolute capacity, or a measured memory ceiling.

For the assignment's human-review requirement, the student must compare these eight corrections with the referenced JTL/JSON and explicitly approve or amend them in `STUDENT_EVIDENCE_REQUIRED.md`. AI cannot truthfully sign that decision.